Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Abstract...

1 Some examples

For internal references use label-refs: see section 1. Bibliographic citations can be done with "cite": refs. [1, 2, 3]. When possible, align equations on the equal sign. The package `amsmath` is already loaded. See eq:x.

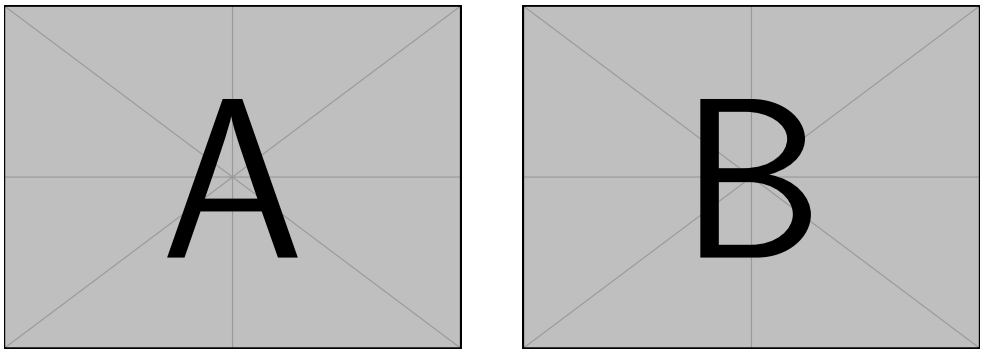
$$x = 1\,, \qquad y = 2\,, z = 3\,. \tag{1}$$

Also, watch out for the punctuation at the end of the equations.

If you want some equations without the tag (number), please use the available starred-environments. For example: $\mathrm{x} = 1$

2 Figures and tables

All figures and tables should be referenced in the text and should be placed on the page where they are first cited or in subsequent pages. Positioning them in the source file after the paragraph where you first reference them usually yield good results. See figure 1 and table 1 for layout examples. Please note that a caption is mandatory and it must be placed at the bottom of both figures and tables.



Gambar 1: Always give a caption.

x	y	x and y
a	b	a and b
1	2	1 and 2
α	β	α and β

Tabel 1: We prefer to have top and bottom borders around the tables.

We discourage the use of inline figures (e.g. `wrapfigure`), as they may be difficult to position if the page layout changes.

We suggest not to abbreviate: “section”, “appendix”, “figure” and “table”, but “eq.” and “ref.” are welcome. Also, please do not use `\emph` or `\it` for latin abbreviaitons: i.e., et al., e.g., vs., etc.

Up to paragraphs. We find that having more levels usually reduces the clarity of the article. Also, we strongly discourage the use of non-numbered sections (e.g. `\subsubsection*`). Please also consider the use of "`\texorpdfstring{}{}`" to avoid warnings from the `hyperref` package when you have math in the section titles.

A Some title

Please always give a title also for appendices.

This is the most common positions for acknowledgments. A macro is available to maintain the same layout and spelling of the heading.

Note added. This is also a good position for notes added after the paper has been written.

Pustaka

[1] Author, *Title*, *J. Abbrev.* **vol** (year) pg.

[2] Author, *Title*, arxiv:1234.5678.

[3] Author, *Title*, Publisher (year).